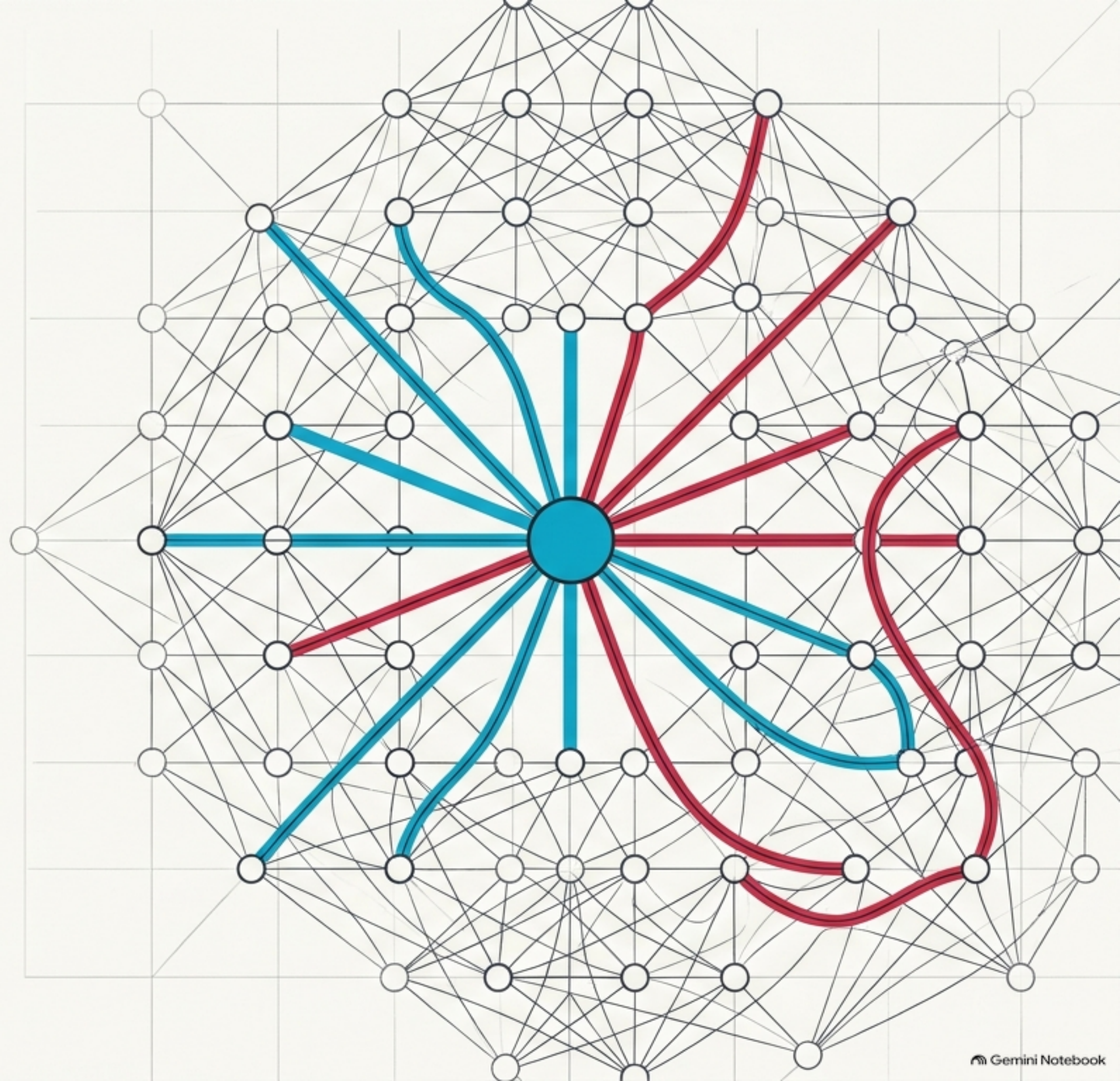


THE CBR MASS GAP CONJECTURE

RELATIONAL COMPUTATION
AND EMERGENT PHYSICS

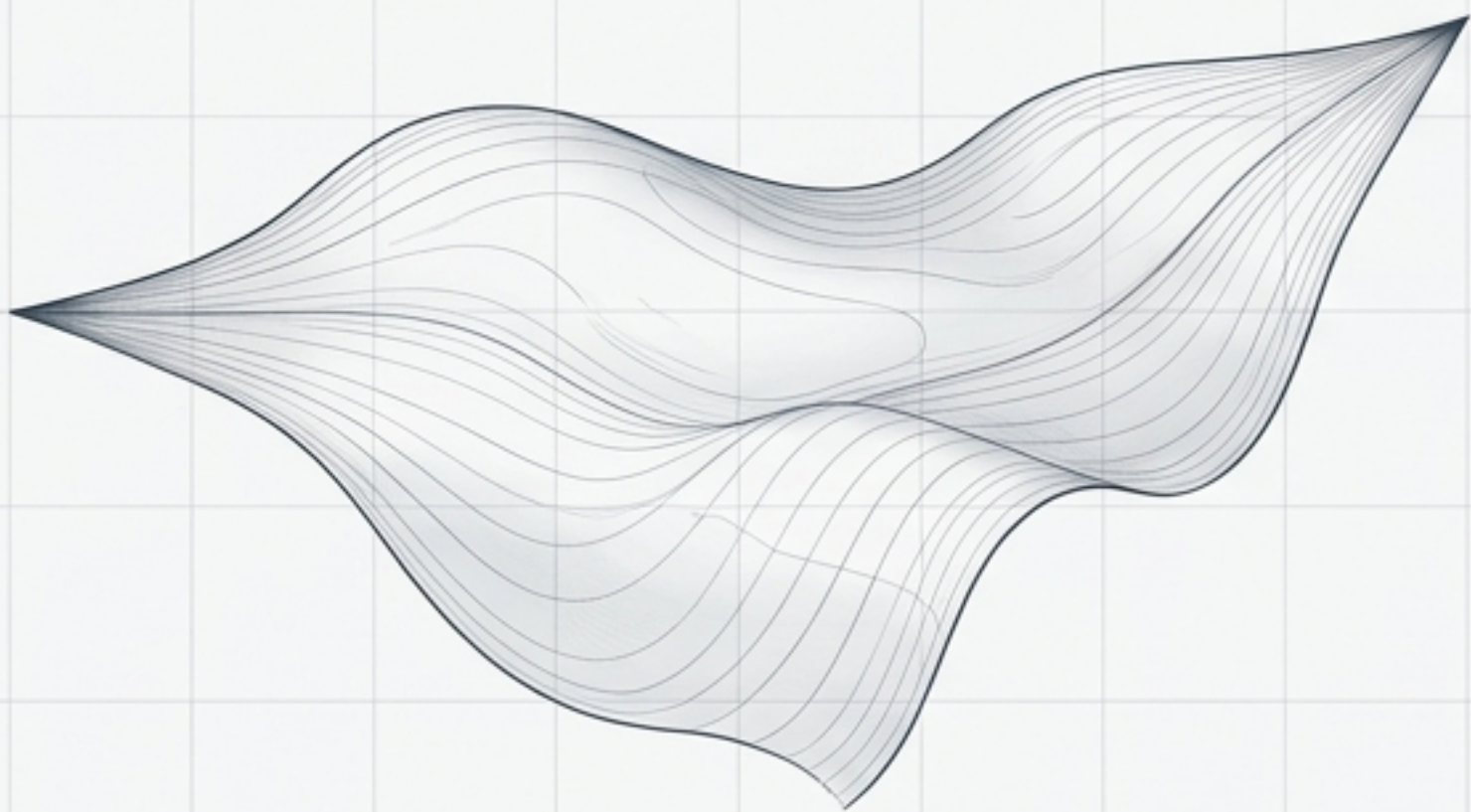
Assume as little
physics as possible.



The Yang-Mills Millennium Problem requires a deeper foundational architecture.

The Problem: The Continuum Assumption

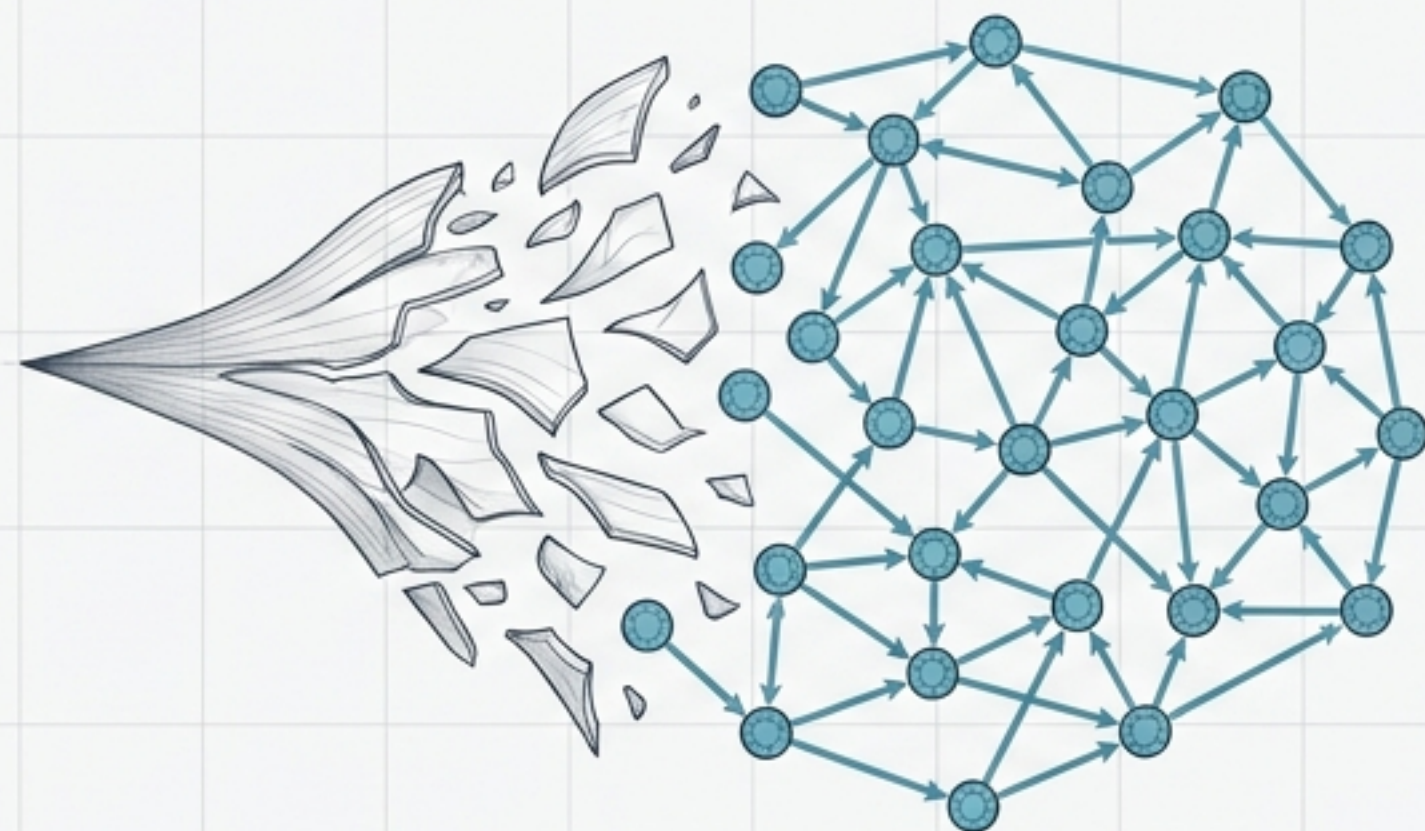
Standard quantum field theory assumes continuous spacetime, fundamental fields, and primitive particles.



Yet, non-Abelian gauge theories exhibit a mysterious "**mass gap**"—a strictly positive minimum excitation cost—that remains unproven mathematically.



The Pivot: Compute-Based Reality (CBR)



Physical reality is generated by locally admissible transformations of relational states subject to global consistency constraints.

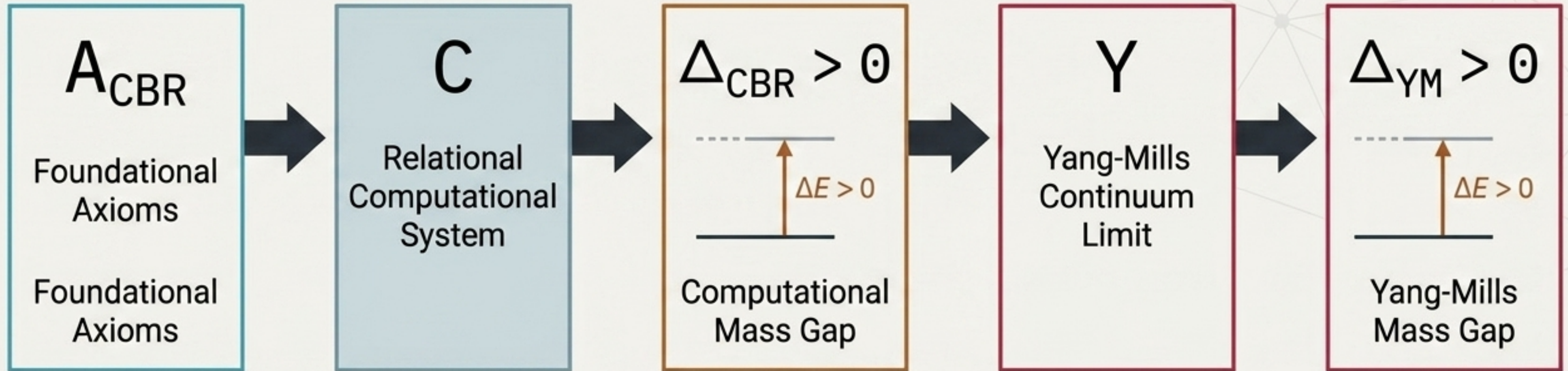
Physics is not an arbitrary set of laws imposed on a pre-existing vacuum.

Rebuilding Physics: The CBR Paradigm Shift

FOUNDATION	Continuous Spacetime	Finite Combinatorial Graphs
FIELDS	Fundamental Entities	Effective Emergent Structures
MASS	Intrinsic Particle Property	{ Irreducible Computational Persistence Cost }

[Rather than assuming continuous topology, we aim to derive continuous spacetime and the Yang-Mills mass gap as effective macroscopic shadows of an underlying computational network.]

The Chain of Implications: Proving the Emergent Gap



A true solution requires proving every single implication. Establishing the computational system without the continuum limit is insufficient.

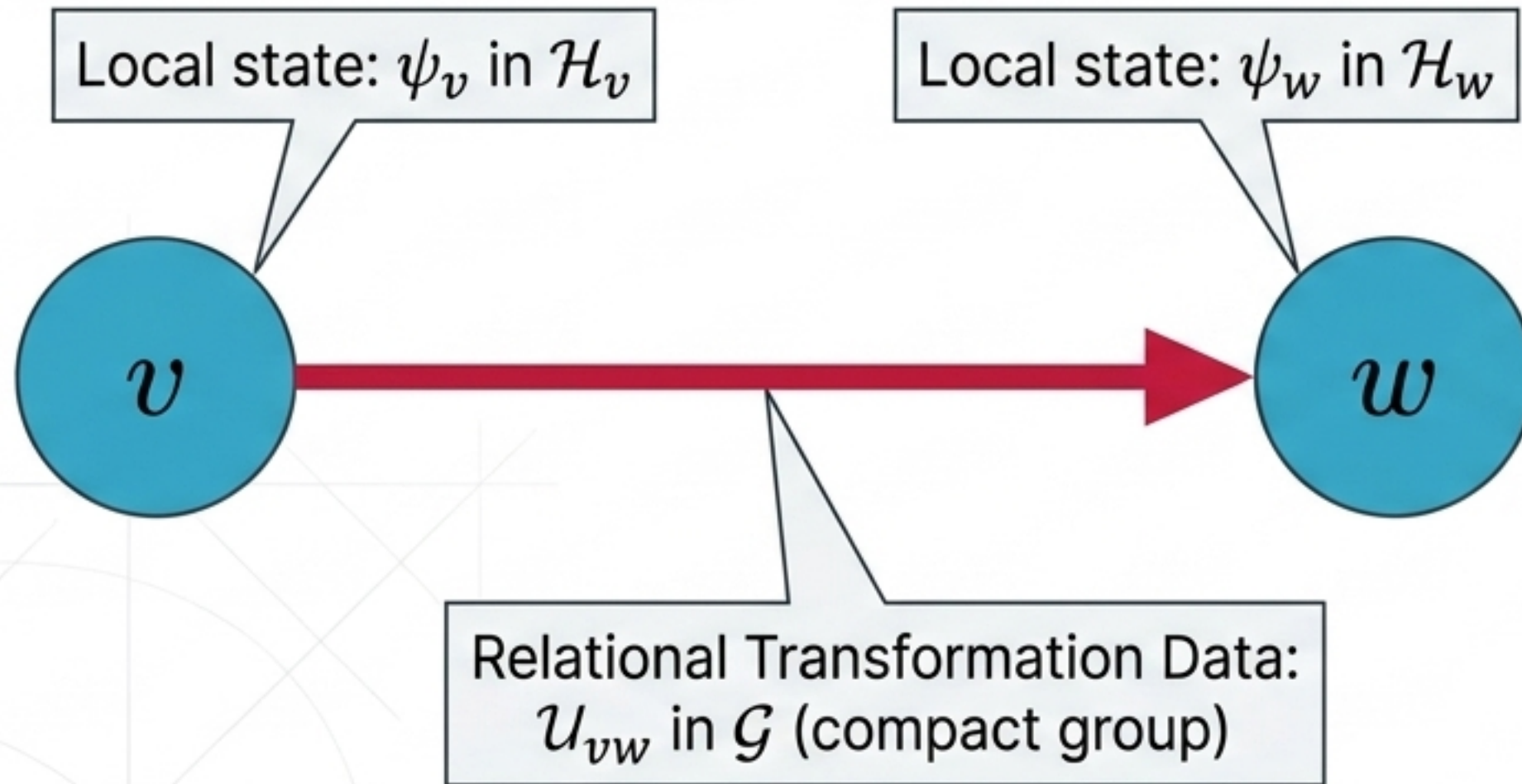
[CBR computational system $\Rightarrow$ positive spectral gap]

AND

[CBR computational system $\Rightarrow$ 4D Yang-Mills continuum limit]

$$A_{\text{CBR}} \rightarrow C \rightarrow \Delta_{\text{CBR}} \rightarrow Y \rightarrow \Delta_{\text{YM}}$$

The Primitive Architecture: Defining the Relational Substrate



The Microscopic State Space

$$X = (K, \mathcal{H}, \Psi, \mathcal{U})$$

K : Finite combinatorial structure (e.g., graphs, simplicial complexes). Topology is not fixed.

$\mathcal{H}$: Total microscopic state space (Tensor product of $\mathcal{H}_v$).

$\Psi, \mathcal{U}$: Total configurations of local states and transformation data.



Gauge Equivalence: Representation vs. Physical Reality

Raw Network A
(Microscopic Representation 1)

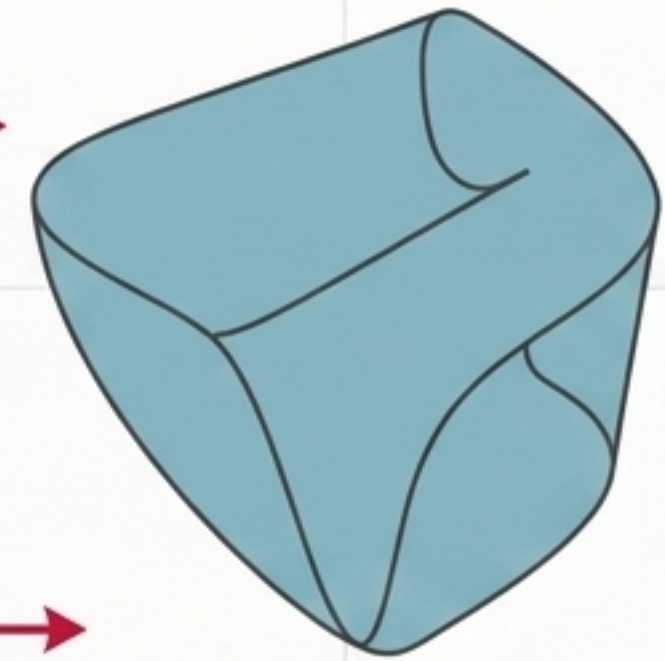


Raw Network B
(Microscopic Representation 2)



$$\psi_v \rightarrow g_v \psi_v$$

Physical Quotient State
($P_K = C_K / G_K$)



Local representation does not constitute physical information.

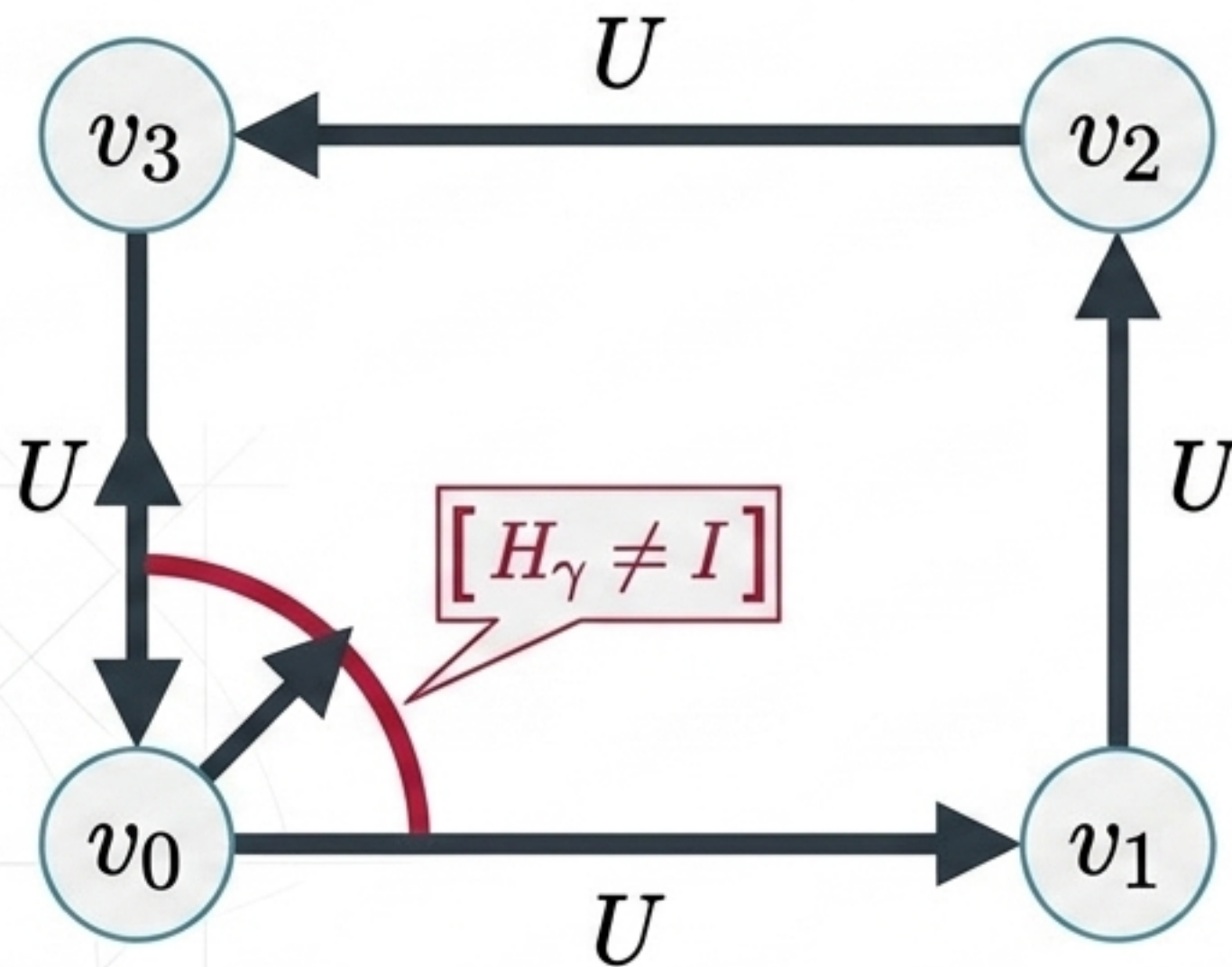
Configurations related by a local transformation g_v in G are physically identical.

[representation is not reality; relational invariants are.]



Relational Dynamics: Curvature as Composition Mismatch

The Holonomy Loop



HOLONOMY (H_γ):

The product of transformations around a closed loop.

THE CBR INTERPRETATION:

If $H_\gamma \neq I$, local relational transformations fail to compose globally into the identity.

CURVATURE:

Field curvature is simply computational composition mismatch.



Computational Energy: Sustaining Reality Costs Work

Computational Cost Functional $C(\Psi)$

αC_{curv} (Curvature: Cost of gauge mismatch)	$\beta C_{\text{constraint}}$ (Constraint: Cost of violating global consistency)	$\gamma C_{\text{persistence}}$ (Persistence: Update work needed to maintain structure)	ηC_{topology} (Topology: Cost of topological obstruction)
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The Vacuum

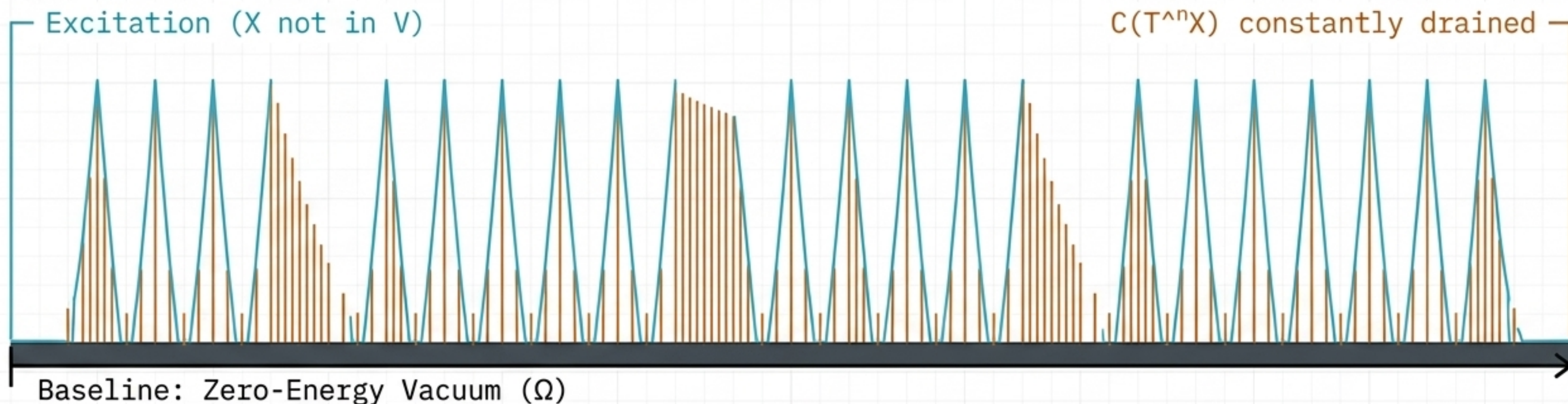
Defined as: $[\Omega] = \text{argmin } [C(\Psi)]$

Normalized so that $[C(\Omega)] = 0.$

The vacuum minimizes the “frustration” of relational mismatch.

A_{CBR} C Δ_{CBR} Y Δ_{YM}

Computational Mass: The Irreducible Persistence Cost

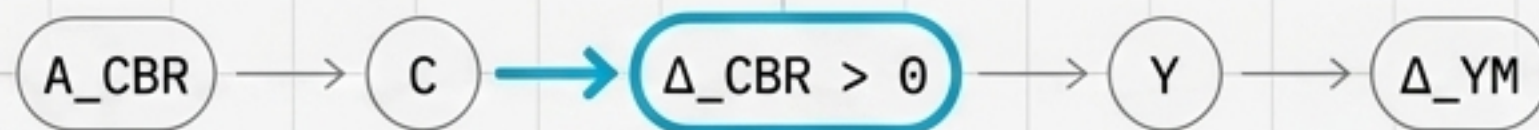


$$X \rightarrow TX \rightarrow T^2X \rightarrow \dots \rightarrow \dots \rightarrow T^NX$$

Burnt Amber

[physical mass $\sim$ irreducible persistence cost]

$$m_C(X) = \liminf_N \frac{1}{N} \sum C(T^n X)$$



The Core Hypothesis: The Computational Mass Gap

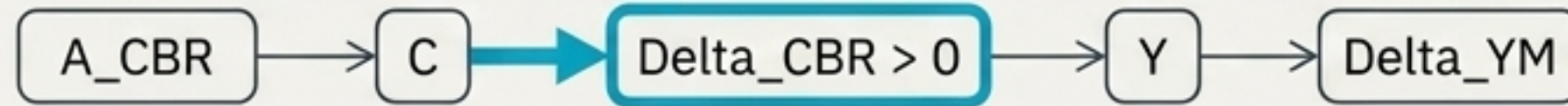
$$\Delta_C = \inf m_C(X)$$

over all X in P_{pers} , X not in V

$$[\Delta_C > 0]$$

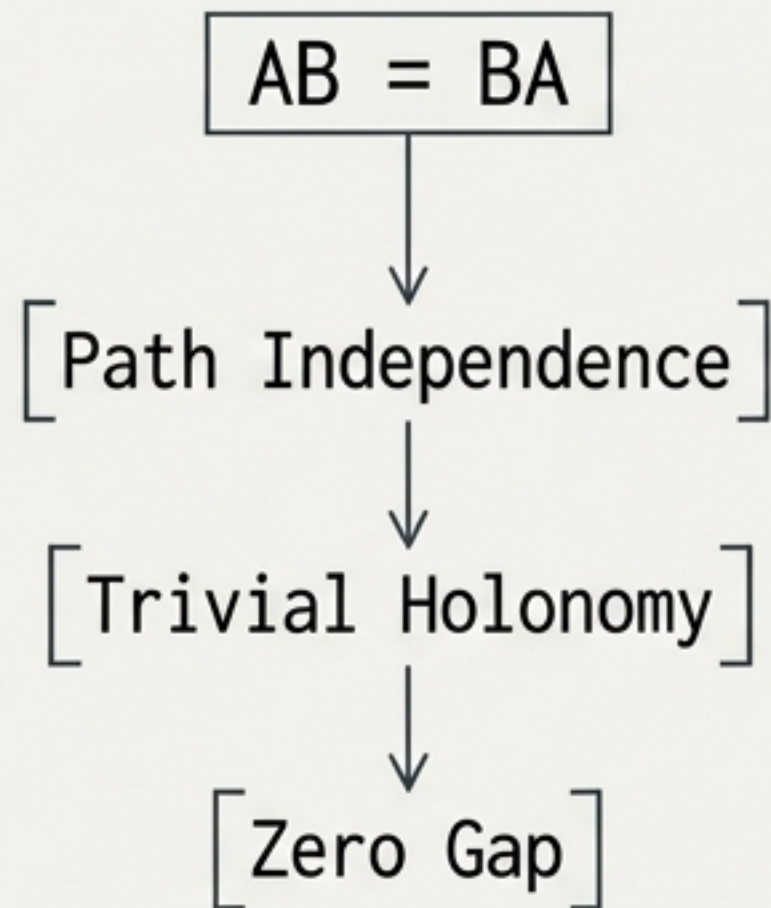
Conjecture 1: For a globally consistent, non-Abelian relational computational system, the minimum computational mass over all persistent gauge-invariant excitations is strictly positive.

Critically: This gap must follow from the structural properties of the system, not merely imposed through discretisation.

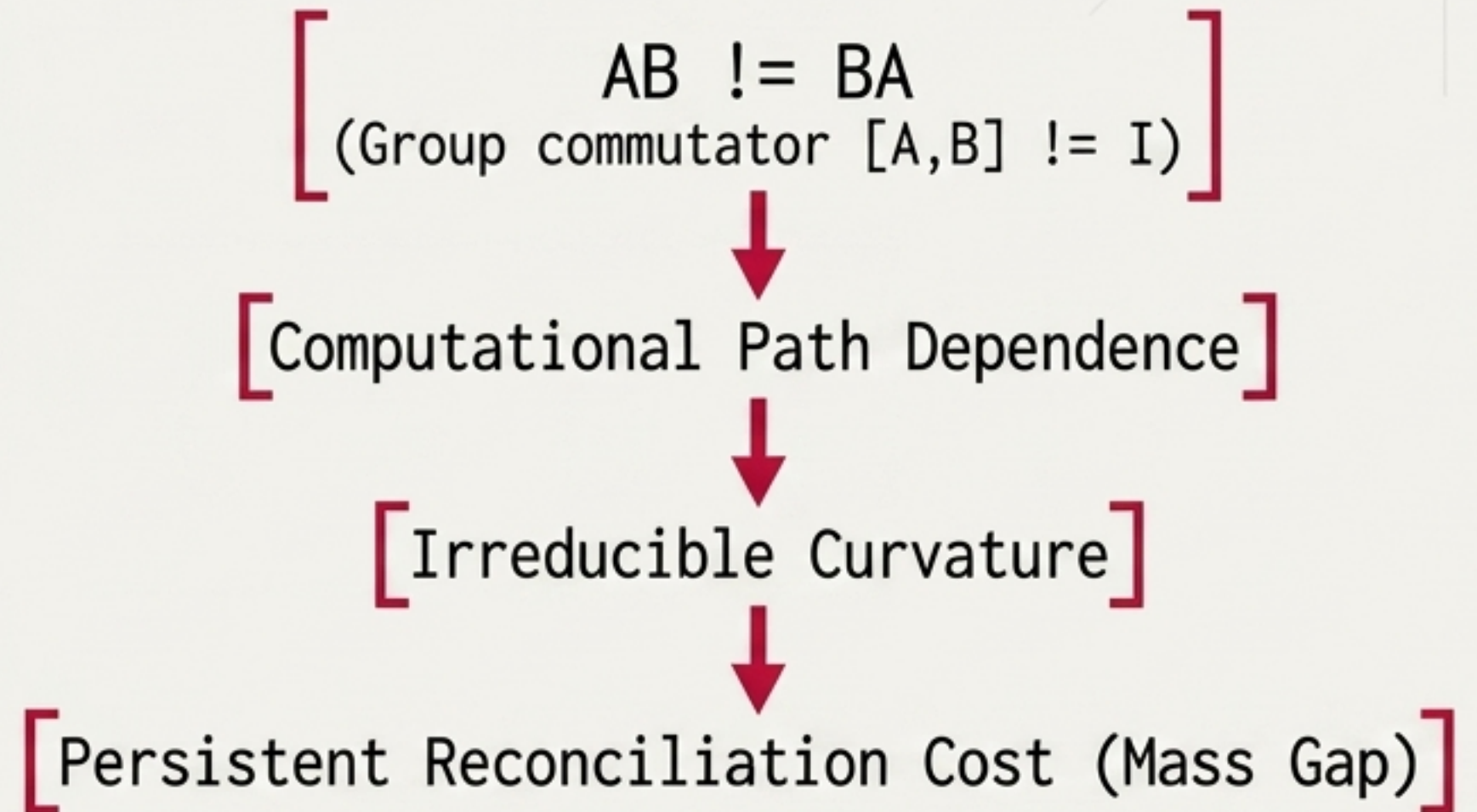


The 'Aha' Moment: Why Non-Abelian Structure is Mandatory

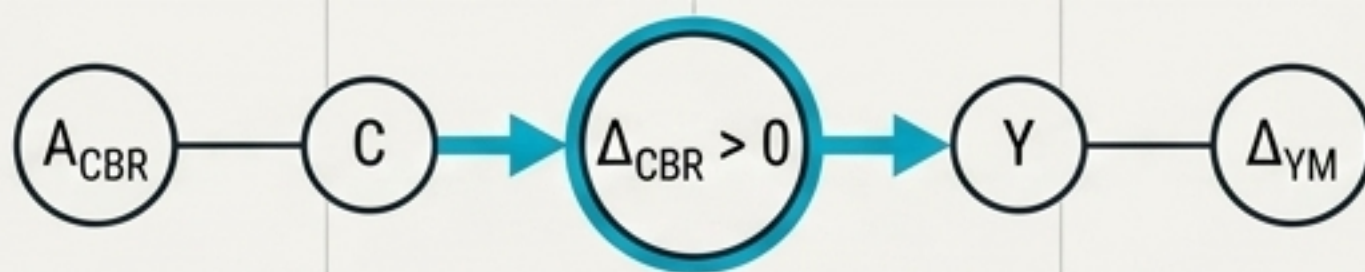
Abelian Dynamics



Non-Abelian Dynamics



Non-commutativity creates an irreducible failure of local transformations to commute, acting as the structural anchor for a positive mass gap.



Six Candidate Mechanisms for the Computational Gap

1. Non-Abelian Frustration

Non-commuting transformations force irreducible curvature.



2. Topological Obstruction

Persistent excitations cannot contract continuously to vacuum (e.g., $C_q > 0$).



3. Constraint Locking

Global consistency creates a minimum non-trivial deviation.



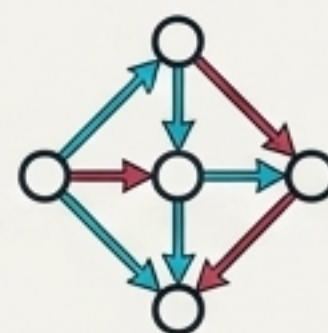
4. Entropic Suppression

Low-energy non-trivial states are dynamically suppressed.



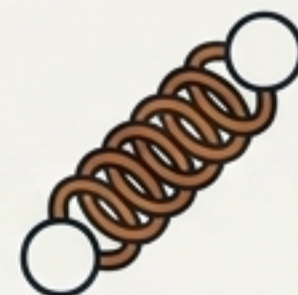
5. Computational Irreducibility

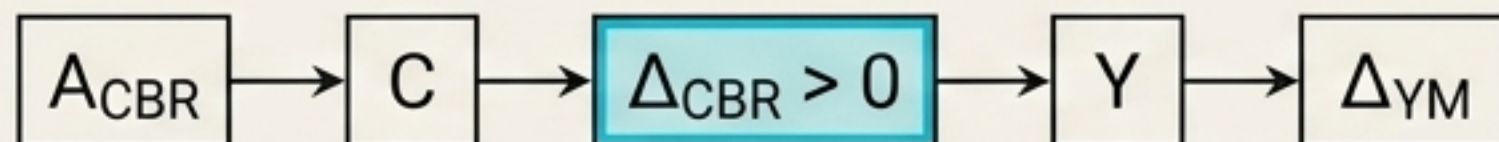
Persistent structure requires a minimum number of dependent transformations.



6. Confinement

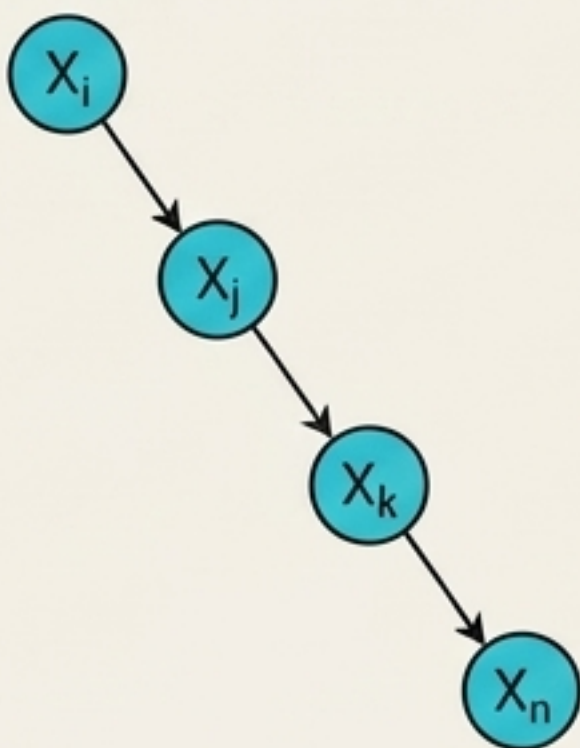
Isolated gauge charges are prohibited, forcing finite-energy composite excitations.





Macroscopic Emergence: Space and Time from Computation

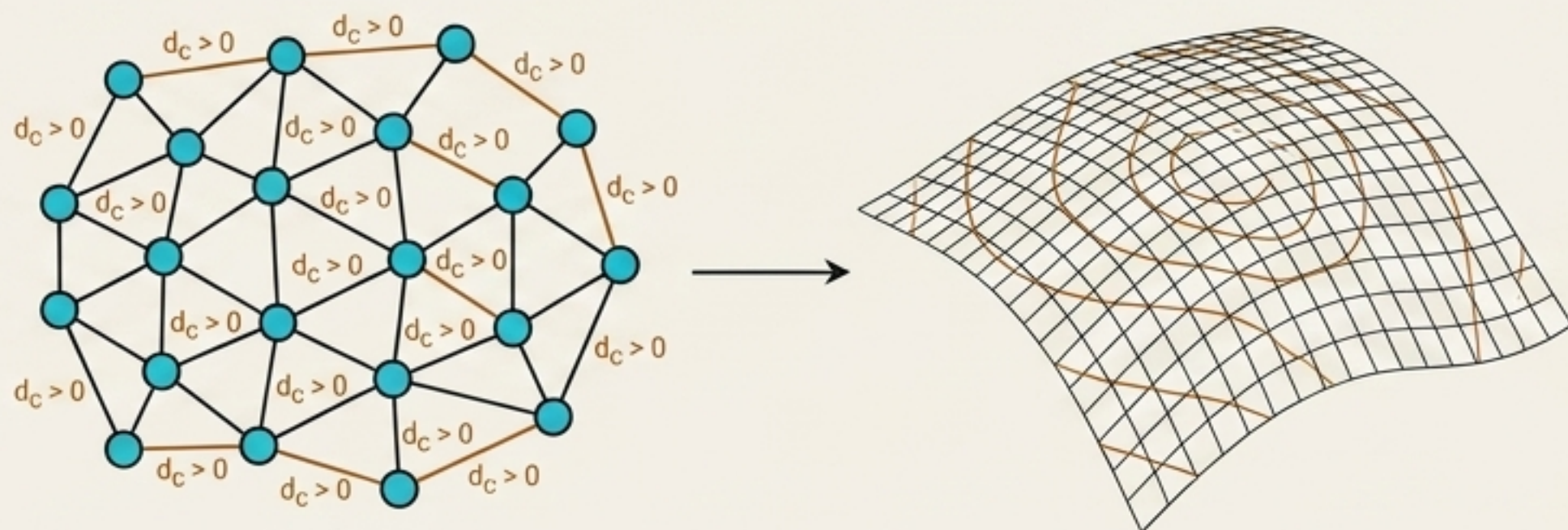
Emergent Time



Computation: Partial ordering of causal dependence ($X_i < X_j$).

[Time = order of state dependence]

Emergent Space



Computation: Computational distance (d_c) based on transition cost.

Continuum Limit: $d_c \rightarrow d_g$ (geodesic distance of emergent metric $g_{\mu\nu}$).

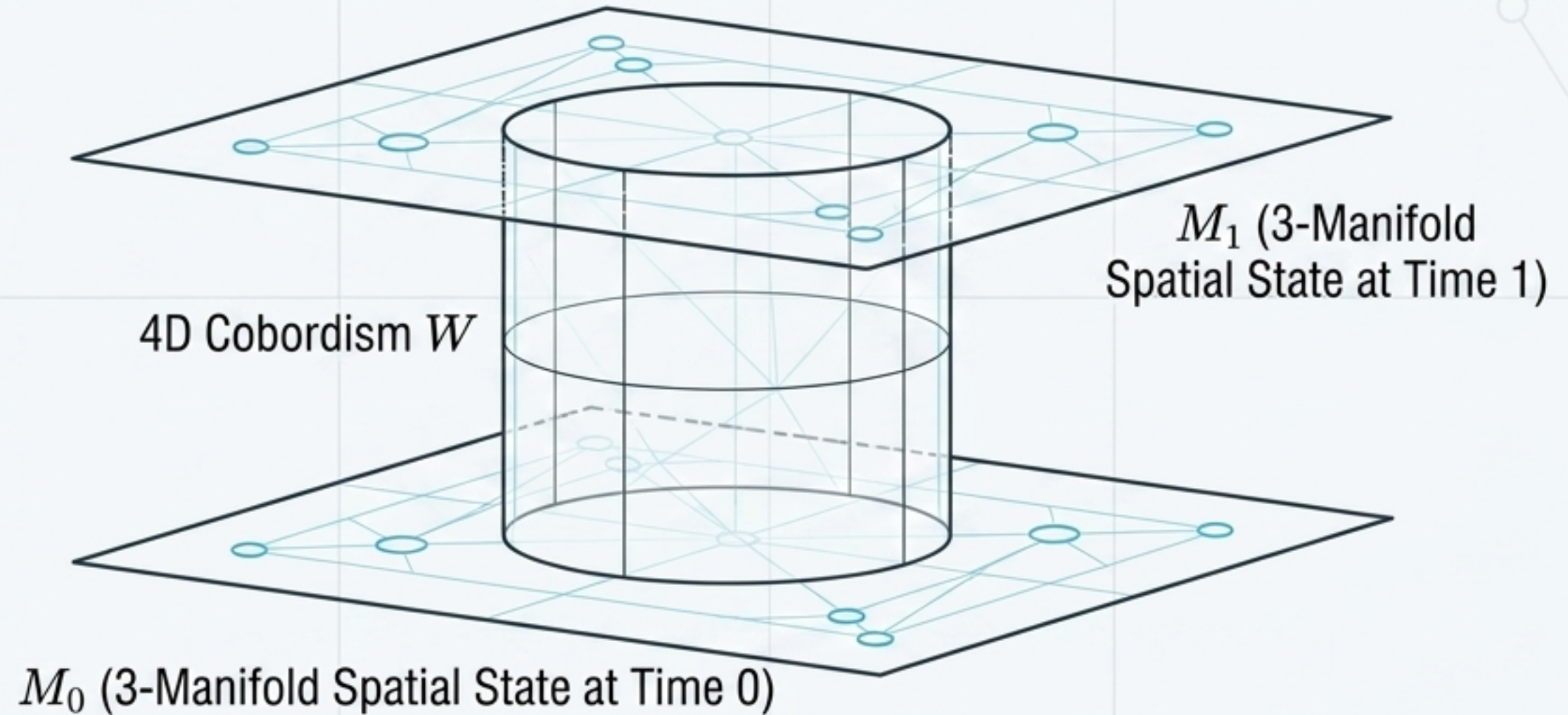
Relativistic Note: Finite computational propagation naturally generates an effective light cone ($d(x,y) \leq vn$), recovering Lorentz symmetry at macroscopic scales.



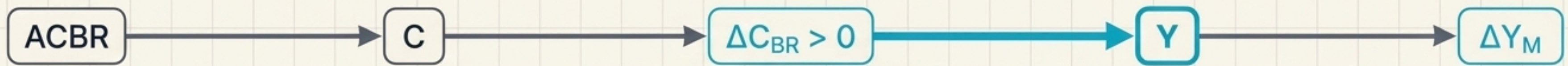
Cobordism as Computation: Generating the 4D History

Take each spatial computational state to be a 3-manifold with relational fields.

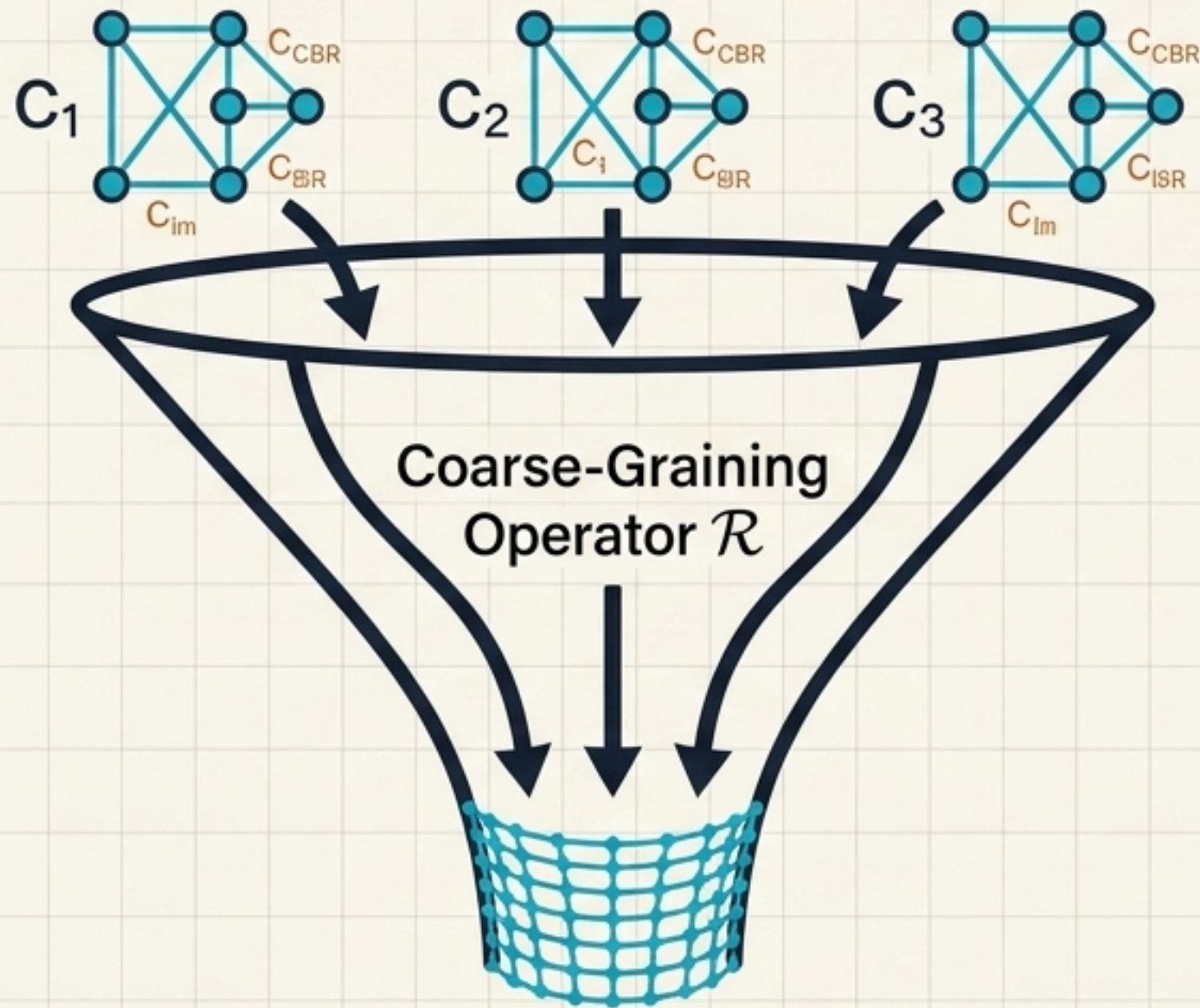
A sequence of transformations naturally generates an effective four-dimensional structure without assuming 4D spacetime a priori.



$\left[\left[\text{3D relational state} + \text{transition} \Rightarrow \text{emergent 4D history} \right] \right]$



The CBR-Yang-Mills Correspondence Conjecture



Universality Class:
4D Quantum Yang-Mills Theory

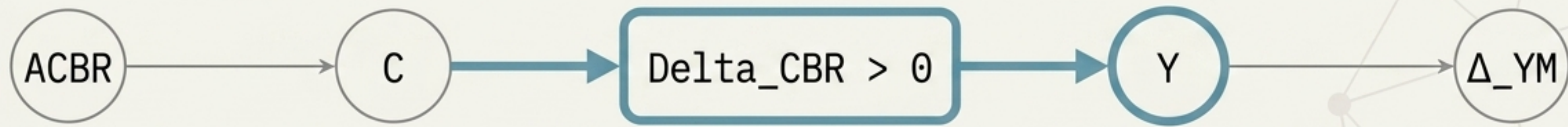
Key Conditions:

- **Continuum Limit:** Microscopic relational scale $a \rightarrow 0$
- **Infinite-Volume Limit:** System size $L \rightarrow \text{infinity}$

Correspondence Axiom:

- Under repeated coarse-graining, the generating functional converges:
 $Z_{\text{CBR}} \rightarrow Z_{\text{YM}}$.

[the microscopic implementation of reality need not be unique]



The Strong Conjecture: Evading the Circularity Trap

Discreteness $\rightarrow$ Gap

Merely asserting that a discrete system has discrete energy levels ($C(X) \geq 1$) does not solve the physical problem. The gap must survive arbitrary refinement, increasing volume, and renormalisation.

The Critical Mathematical Obstacle

$$\liminf_{a \rightarrow 0} \liminf_{L \rightarrow \infty} \Delta_C(a, L) = \Delta_* > 0$$

[microscopic discreteness $\neq$ physical mass gap]

The Dual-Track Research Engine

Track 1: Experimental Discovery (Simulation)



Tools: Spectral graph theory, Tensor Networks, Finite systems simulations (e.g., $SU(2)$ on triangulated 3-manifolds).

Goal: Identifies candidate structures and estimates $\Delta_C(L) \rightarrow ?$ as $L \rightarrow \infty$.

Track 2: Rigorous Proof (Mathematics)



Tools: Uniform bounds, Representation theory, Information geometry.

Goal: Establishes necessity. Proves continuum limits and gap preservation.

Key Principle: Simulation discovers; mathematics proves. Proof and experiment must remain rigorously separate.

The Ascent to Yang-Mills: 11 Mathematical Milestones

The Peak (Millennium Problem)

M10: YM Correspondence

M11: Gap Preservation
[$\Delta_{YM} > 0$]

The Summit Approach (Continuum)

M9: Continuum limit [$a \rightarrow 0$]

The Ridge (Finite Systems)

M8: Renormalisation

M7: Uniform Bound

M6: Finite Gap [$\Delta_c(L) > 0$]

The Valley (Foundations)

M5: Persistent Sector

M4: Vacuum

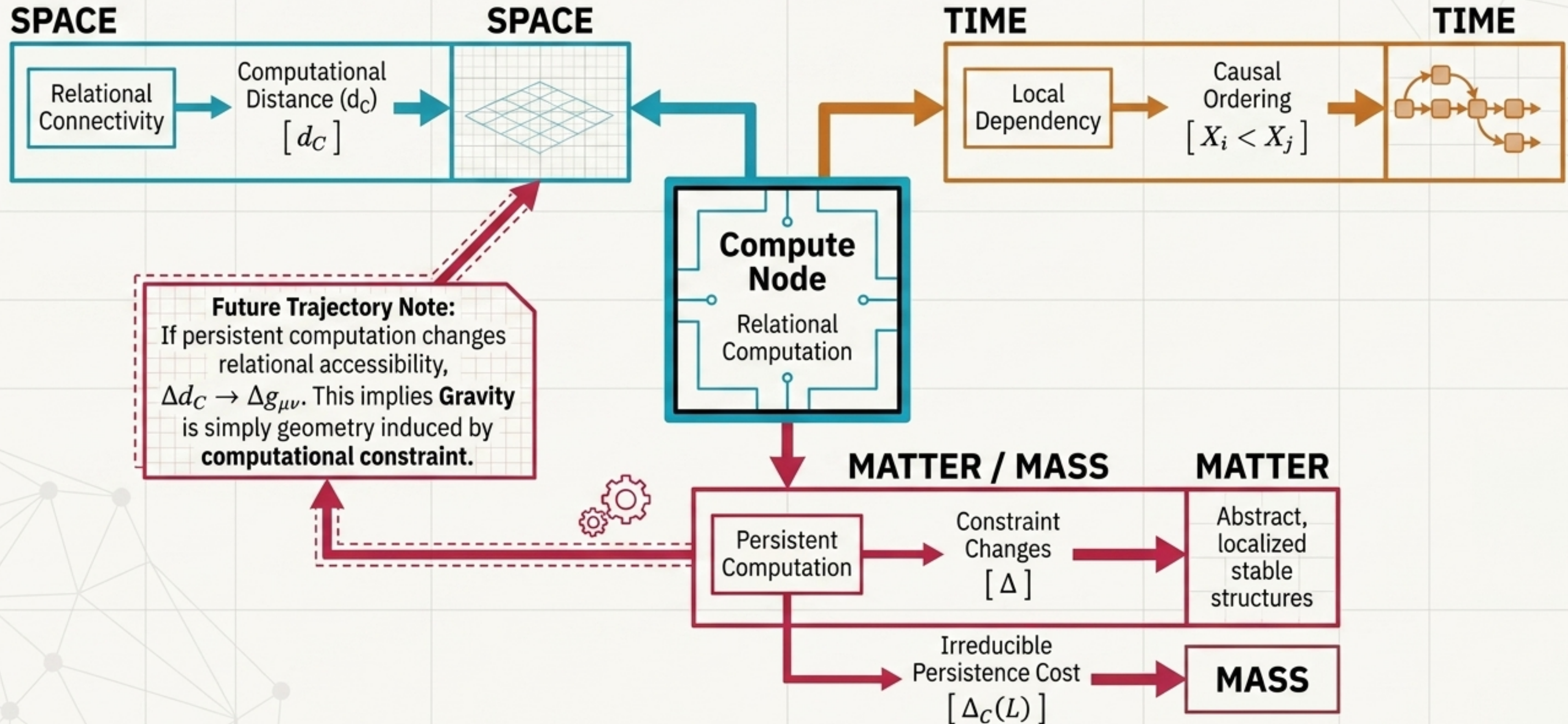
M3: Dynamics

M2: Gauge Quotient

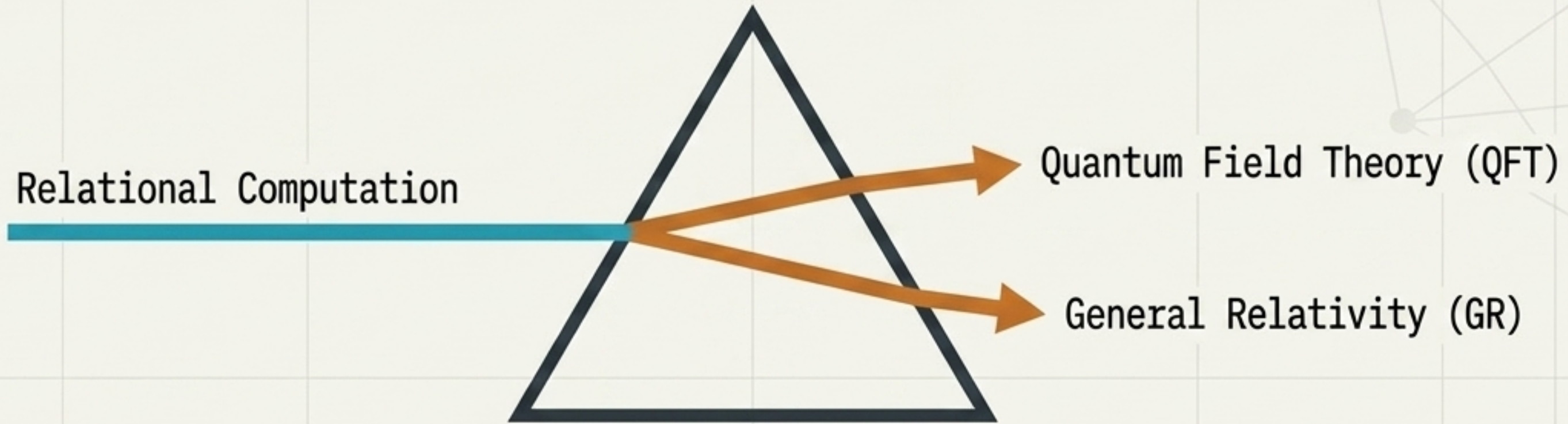
M1: State Space

Note: Each milestone is an independently meaningful research problem that moves CBR from a conceptual framework to a productive mathematical programme.

Synthesis: The Emergence of Spacetime and Matter



Toward Universal Physics: The Stable Invariants of Computation



[Assume as little physics as possible.]

Physical Law as Invariants of Computation

Physics is not an additional collection of laws imposed upon a system. Physical law is the structure of the stable invariants of computation.

The Yang-Mills Mass Gap as Persistence Cost

The Yang-Mills mass gap is no longer just a property of a quantum field—it is the ultimate consequence of what persistent existence costs.